

Darb Public Transit Management System



System Overview

Darb is a centralized relational database architecture designed to manage urban public transportation operations, fare collection, and passenger journeys. The system links passenger accounts (Users) with contactless transit cards (Cards), enabling automated multi-modal trip tracking across city routes (Routes), transport vehicles (Vehicles), and station checkpoints (Stations).

Furthermore, the schema supports operational efficiency by tracking active personnel assignments (Drivers) and managing flexible fare products (Tickets). Built with relational constraints, primary/foreign key mappings, and domain-level attribute checks, Darb ensures data integrity across real-time ticketing, journey logging, and fleet management.

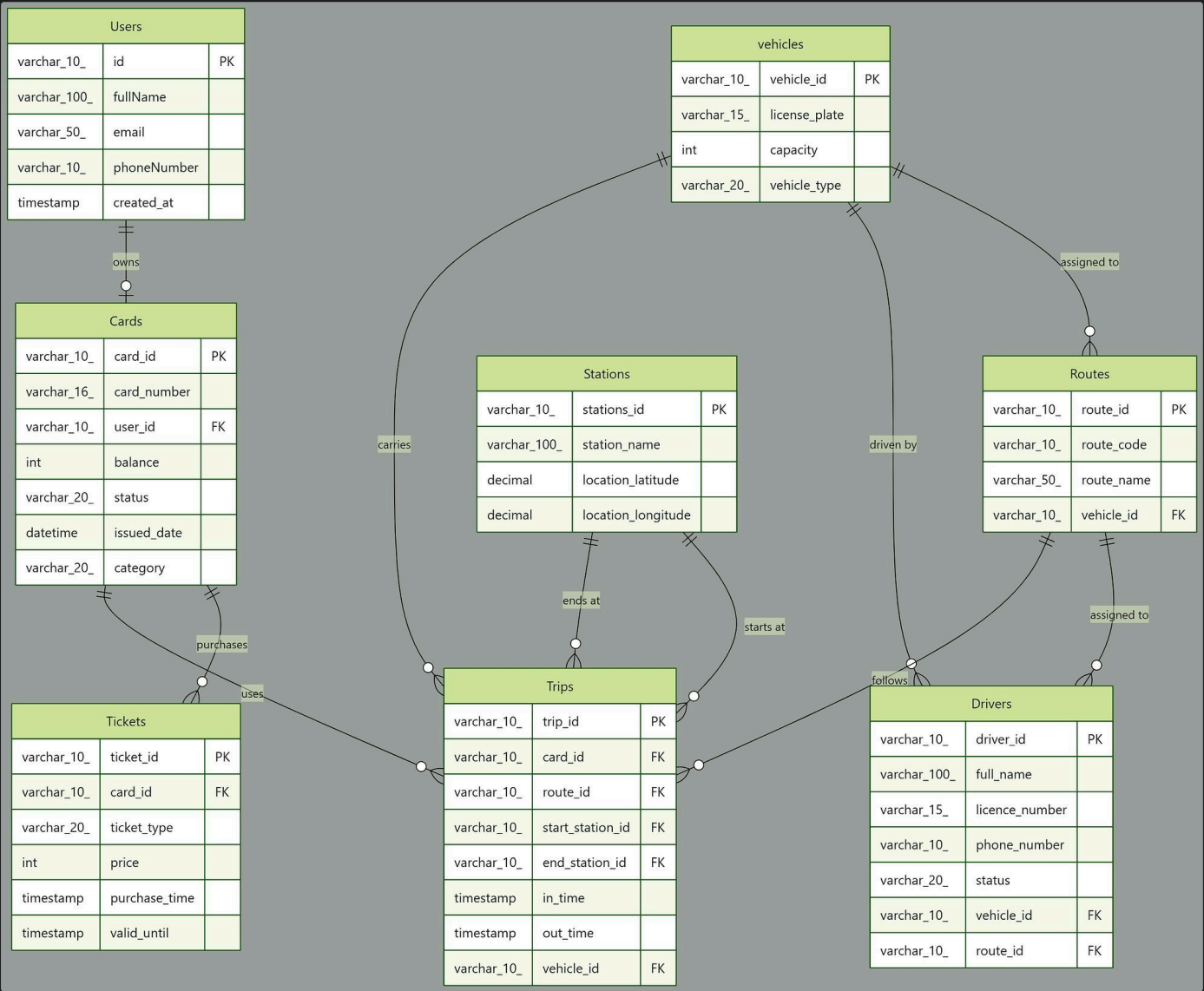


Table 1: Users

Primary Key: id (varchar(10))

Column	Type	Constraints
id	varchar(10)	PRIMARY KEY
fullName	varchar(100)	UNIQUE, NOT NULL
email	varchar(50)	UNIQUE, NOT NULL
phoneNumber	varchar(10)	UNIQUE, NOT NULL
created_at	timestamp	DEFAULT CURRENT_TIMESTAMP

Purpose

Stores passenger account information and contact details for all registered system users.

Key Design Notes

- All contact fields enforced as unique to prevent duplicate accounts
- Automatic timestamp on account creation
- Serves as the root entity linked to Cards via 1:1 relationship

Table 2: Cards

Primary Key: card_id (varchar(10)) | **Foreign Key:** user_id → Users(id) — 1:1 relationship

Column	Type	Constraints / Values
card_id	varchar(10)	PRIMARY KEY
card_number	varchar(16)	UNIQUE, NOT NULL
user_id	varchar(10)	UNIQUE, NOT NULL, FK → Users
balance	int	DEFAULT 0, CHECK ≥ 0
status	varchar(20)	DEFAULT 'Active' Suspended Expired
issued_date	datetime	NOT NULL
category	varchar(20)	standard VIP

Card Status
Active · Suspended · Expired

Category Tier
Standard · VIP

Balance Guard
CHECK constraint prevents negative balance

Purpose: Manages contactless transit cards linked to users for fare payment and trip tracking.

Tables 3 & 4: Stations & Vehicles

TABLE 3

Stations

Primary Key: stations_id (varchar(10))

Column	Type	Constraints
stations_id	varchar(10)	PRIMARY KEY
station_name	varchar(100)	UNIQUE, NOT NULL
location_latitude	decimal(10, 8)	NOT NULL
location_longitude	decimal(10, 8)	NOT NULL

Defines transit stations with geographic coordinates for route mapping and trip endpoints.

TABLE 4

Vehicles

Primary Key: vehicle_id (varchar(10))

Column	Type	Constraints
vehicle_id	varchar(10)	PRIMARY KEY
license_plate	varchar(15)	UNIQUE, NOT NULL
capacity	int	CHECK capacity > 0
vehicle_type	varchar(20)	Bus Metro Bus on demand

Tracks fleet vehicles with capacity and type information for route assignment and trip logging.

Tables 5 & 6: Routes & Drivers

TABLE 5

Routes

Primary Key: route_id (varchar(10))
Foreign Key: vehicle_id → Vehicles(vehicle_id)

Column	Notes
route_id	PRIMARY KEY
vehicle_id	FK → Vehicles
route_code	Route identifier code
route_name	Descriptive route name

TABLE 6

Drivers

Primary Key: driver_id (varchar(10))
Foreign Keys: vehicle_id → Vehicles, route_id → Routes

Column	Notes
driver_id	PRIMARY KEY
full_name	Driver full name
licence_number	Unique licence ID
phone_number	Contact number
status	Active On_Leave Suspended



Route Assignment

Each route is bound to a specific vehicle, ensuring fleet accountability across all city corridors.



Driver Status Tracking

Three-state status (Active, On Leave, Suspended) enables real-time personnel management and scheduling.



Dual Foreign Keys

Drivers reference both Vehicles and Routes, creating a complete operational assignment chain.

Tables 7 & 8: Tickets & Trips

TABLE 7

Tickets

Primary Key: ticket_id (varchar(10))

Foreign Key: card_id → Cards(card_id)

Column	Notes
ticket_id	PRIMARY KEY
card_id	FK → Cards
ticket_type	2 Hours 3 days 7 days 30 days
price	Fare amount
purchase_time	Timestamp of purchase
valid_until	Expiry datetime

TABLE 8

Trips

Primary Key: trip_id (varchar(10))

Foreign Keys: card_id, route_id, start_station_id, end_station_id, vehicle_id

Column	Notes
trip_id	PRIMARY KEY
card_id	FK → Cards
route_id	FK → Routes
start_station_id	FK → Stations
end_station_id	FK → Stations
vehicle_id	FK → Vehicles
in_time	Boarding timestamp
out_time	Alighting timestamp

-  The Trips table is the most connected entity in the schema — it references 5 foreign keys, serving as the central fact table that ties together passengers, routes, stations, and vehicles into a complete journey record.

Database Creation Script

Below is the complete SQL script for creating the Darb database and all its associated tables, including primary keys, foreign keys, unique constraints, NOT NULL constraints, default values, and check constraints to ensure data integrity.

```
CREATE DATABASE DarbDB;

-- Table 1: Users
CREATE Table Users (
  id varchar(10) primary key ,
  fullName varchar(100) unique not null ,
  email varchar(50) not null unique ,
  phoneNumber varchar(10) unique not null ,
  created_at timestamp default current_timestamp
);

-- Table 2: Cards
CREATE Table Cards (
  card_id varchar(10) primary key ,
  card_number varchar(16) not null unique ,
  user_id varchar(10) not null unique ,
  foreign key (user_id) references Users(id) ,
  balance int default 0 check (balance >=0) ,
  status varchar(20) default 'Active' check (status = 'Active' or status = 'Suspended' or status = 'Expired'),
  issued_date datetime not null ,
  category varchar(20) not null check (category = 'standard' or category = 'VIP')
);

-- Table 3: Stations
CREATE Table Stations (
  stations_id varchar(10) primary key ,
  station_name varchar(100) not null unique ,
  location_latitude decimal(10,8) not null ,
  location_longitude decimal(10, 8) not null
);

-- Table 4: Vehicles
CREATE Table vehicles (
  vehicle_id varchar(10) primary key ,
  license_plate varchar(15) not null unique ,
  capacity int not null check (capacity>0),
  vehicle_type varchar(20) not null check (vehicle_type = 'Bus' or vehicle_type = 'Metro' or vehicle_type = 'Bus on demand')
);

-- Table 5: Routes
CREATE Table Routes (
  route_id varchar(10) primary key ,
  route_code varchar(10) not null unique ,
  route_name varchar(50) not null ,
  vehicle_id varchar(10) not null,
  foreign key (vehicle_id) references vehicles(vehicle_id)
);

-- Table 6: Drivers
CREATE Table Drivers (
  driver_id varchar(10) primary key ,
  full_name varchar(100) not null unique ,
  licence_number varchar(15) not null unique ,
  phone_number varchar(10) not null unique ,
  status VARCHAR(20) not null DEFAULT 'Active' CHECK (status ='Active' or status = 'On_Leave' or status = 'Suspended'),
  vehicle_id varchar(10) not null ,
  foreign key (vehicle_id) references vehicles(vehicle_id),
  route_id varchar(10) references Routes(route_id)
);

-- Table 7: Tickets
CREATE Table Tickets (
  ticket_id varchar(10) primary key ,
  card_id varchar(10) not null ,
  foreign key (card_id) references Cards(card_id),
  ticket_type varchar(20) not null check (ticket_type = '2 Hours' or ticket_type = '3 days' or ticket_type = '7 days' or ticket_type = '30 days'),
  price int not null check (price > 0),
  purchase_time timestamp default current_timestamp not null ,
  valid_until timestamp default current_timestamp not null
);

-- Table 8: Trips
CREATE Table Trips (
  trip_id varchar(10) primary key ,
  card_id varchar(10) not null ,
  foreign key (card_id) references Cards(card_id),
  route_id varchar(10) not null ,
  foreign key (route_id) references Routes(route_id),
  start_station_id varchar(10) not null ,
  foreign key (start_station_id) references Stations(stations_id) ,
  end_station_id varchar(10) not null ,
  foreign key (end_station_id) references Stations(stations_id),
  in_time timestamp default current_timestamp not null ,
  out_time timestamp default current_timestamp not null ,
  vehicle_id varchar(10) not null ,
  foreign key (vehicle_id) references vehicles(vehicle_id)
);
```